

Monte Carlo

Lecture 2

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- Monte Carlo in the NVT ensemble
- Some practicalities
- Ergodicity
- Ensemble averages
- Monte Carlo in the isobaric-isothermal ensemble
- Monte Carlo in the grand canonical ensemble

Monte Carlo at constant NVT – Canonical Ensemble

Topics covered:

- Monte Carlo in the NVT ensemble
- Practicalities
- Ergodicity
- Ensemble averages
- Monte Carlo in the isobaric-isothermal ensemble
- Monte Carlo in the grand canonical ensemble

The canonical ensemble

N, V, T are all fixed.

Atoms interact; positions fluctuate; energy fluctuates.

We sample configuration space with the **Boltzmann** distribution.

An aside: probabilities, partition functions and ensembles

Partition function for a discrete set of states:

$$Z = \sum_i e^{-E_i/k_B T}$$

Probability of state i :

$$\mathcal{P}_i = \frac{e^{-E_i/k_B T}}{Z}$$

Connection to thermodynamics:

$$F = -k_B T \ln Z$$

$$S = -\left(\frac{\partial F}{\partial T}\right)_V, \quad P = -\left(\frac{\partial F}{\partial V}\right)_T$$

Thermodynamic average

$$\begin{aligned}\langle A \rangle &= \sum_i A_i \mathcal{P}_i \\ &= \frac{\sum_i A_i e^{-E_i/k_B T}}{Z}\end{aligned}$$

Key point

Partition functions can be derived for *any* ensemble (NVT, NPT, μ VT). Each gives the same thermodynamics in the thermodynamic limit.

The NVT (canonical) partition function

For a classical system of N identical atoms:

$$Q_{NVT} = \frac{1}{N! h^{3N}} \int e^{-\mathcal{H}(\mathbf{r}^N, \mathbf{p}^N)/k_B T} d\mathbf{p}^N$$

Separating kinetic and potential parts:

$$Q_{NVT} = \frac{1}{N! \Lambda^{3N}} \underbrace{\int e^{-U(\mathbf{r}^N)/k_B T} d\mathbf{r}^N}_Z$$

where $\Lambda = h/\sqrt{2\pi m k_B T}$ is the thermal de Broglie wavelength, and Z is the **configurational integral**.

What is held constant?

Fixed	Fluctuates
N, V, T	U, P, μ

Monte Carlo samples $\mathbf{r}^N$ with weight $e^{-U/k_B T}$ — we *never* need to evaluate the full integral Z explicitly.

Thermodynamic averaging

The basic equations for the canonical ensemble:

$$\langle A \rangle = \frac{\int A(\mathbf{r}^N) e^{-U(\mathbf{r}^N)/k_B T} d\mathbf{r}^N}{\int e^{-U(\mathbf{r}^N)/k_B T} d\mathbf{r}^N}$$

For example, the average energy $\langle E \rangle$:

$$\langle E \rangle = \frac{3}{2} N k_B T + \frac{\int U(\mathbf{r}^N) e^{-U/k_B T} d\mathbf{r}^N}{\int e^{-U/k_B T} d\mathbf{r}^N} = \frac{3}{2} N k_B T + \langle U \rangle$$

Key message

So we can summarise the basic equations used in the canonical ensemble to calculate thermodynamic properties of a closed, isothermal system. We can of course derive partition functions appropriate for other ensembles.

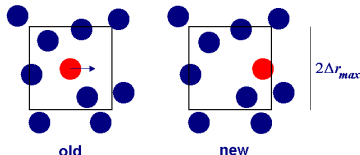
Metropolis Monte Carlo

in the NVT Ensemble

Particle moves

- Pick particle **at random**
(required to obey detailed balance)
- Alter x , y and z coordinates by displacements drawn randomly from:

$$-\Delta r_{\max} < \delta r_{\alpha} < +\Delta r_{\max}$$



Selected particle (red) displaced within a cube of side $2\Delta r_{\max}$

Metropolis criterion

Accept the move with probability:

$$P_{\text{acc}} = \min\left(1, e^{-\Delta U/k_{\text{B}}T}\right)$$

Metropolis acceptance criterion

- If energy of new configuration is **lower** than its predecessor $\Rightarrow$ **accept** the new configuration.
- If energy is **higher** $\Rightarrow$ calculate Boltzmann factor $e^{-\Delta U/k_B T}$ of the energy difference.
 - Generate random number $\xi \in [0, 1]$.
 - If $\xi > e^{-\Delta U/k_B T}$: **reject**; keep original configuration.
 - If $\xi < e^{-\Delta U/k_B T}$: **accept**; new configuration becomes next state.
- Thus *some* moves are to states of **higher energy**.
- The smaller the uphill move, the greater the probability of acceptance.

Why it works

The Metropolis algorithm satisfies **detailed balance**:

$$\mathcal{P}(\mathbf{r}) T(\mathbf{r} \rightarrow \mathbf{r}') = \mathcal{P}(\mathbf{r}') T(\mathbf{r}' \rightarrow \mathbf{r})$$

For the Boltzmann distribution $\mathcal{P} \propto e^{-U/k_B T}$ this requires:

$$\frac{T(\mathbf{r} \rightarrow \mathbf{r}')}{T(\mathbf{r}' \rightarrow \mathbf{r})} = e^{-\Delta U/k_B T}$$

The criterion

$P_{\text{acc}} = \min(1, e^{-\Delta U/k_B T})$ satisfies this exactly.

Detailed balance is sufficient but not necessary — it is however the simplest condition to enforce in practice.

Consequences

- Long runs sample configurations proportional to $e^{-U/k_B T}$
- Ensemble averages converge to correct thermodynamic values
- *Ergodicity* is also required: all of configuration space must be reachable

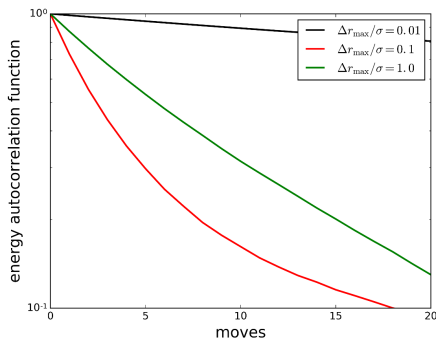
Tuning the acceptance rate

- Acceptance rate is controlled by the maximum displacement $\Delta r_{\max}$.
- Displacements result in an energy change ΔU .
- **Large** displacements are unlikely to be accepted.
- **Small** displacements are more likely accepted, but progress through configuration space is slow.
- Trial displacements should be adjusted to give **maximum displacement per unit CPU time**.

Practical guideline

Tune $\Delta r_{\max}$ to achieve $\approx 50\%$ acceptance ratio. This minimises the statistical inefficiency (correlation time) of the simulation.

Effect of maximum displacement on sampling



The **intermediate** maximum trial move distance ($\Delta r_{\max}/\sigma = 0.1$, red curve) gives the fastest decay in the energy autocorrelation function — the fastest sampling of phase space.

- **Too small** (0.01σ , black): always accepted, but tiny steps $\Rightarrow$ slow decorrelation.
- **Too large** (1.0σ , green): most moves rejected $\Rightarrow$ slow decorrelation.
- **Optimum** (0.1σ): fastest exploration of phase space.

Equilibration

- Monte Carlo simulations converge until equilibrium (or “thermalise”).
- Starting from a non-equilibrium configuration, deviation from equilibrium decreases as $\exp(-t/\tau_0)$, where τ_0 is the **correlation time**.
- Only retain measurements for ensemble averages after $t \gg \tau_0$ — the initial (often unphysical) configuration must be forgotten.

Practical approach

Monitor running averages of U , P , and other key properties as a function of MC steps. When these stabilise around a steady value, equilibration is complete.

See the exercises and the molecular dynamics simulation lectures for more details.

Key quantities computed at each step and averaged:

Energy

$$\langle U \rangle = \frac{1}{M} \sum_{k=1}^M U_k$$

Pressure (virial route)

$$P = \frac{Nk_B T}{V} + \frac{\langle W \rangle}{3V}$$

Isochoric heat capacity

$$\frac{C_V}{Nk_B} = \frac{3}{2} + \frac{\langle U^2 \rangle - \langle U \rangle^2}{N(k_B T)^2}$$

Activity (Widom insertion)

$$z = \frac{1}{\langle e^{-\Delta U_{\text{test}}/k_B T} \rangle}$$

Configurational temperature

$$T_{\text{conf}} = \frac{\langle |\nabla U|^2 \rangle}{k_B \langle \nabla^2 U \rangle}$$

Note

All accumulated during a single NVT run at negligible extra cost.

Configurational Temperature

The **configurational temperature** is defined purely from the potential energy surface:

$$T_{\text{conf}} = \frac{\langle |\nabla U|^2 \rangle}{k_B \langle \nabla^2 U \rangle}$$

- At equilibrium: $T_{\text{conf}} = T_{\text{input}}$
- Requires computing forces and their divergence
- A sensitive test of:
 - Programming correctness
 - Whether detailed balance is truly satisfied
 - Whether equilibration is complete

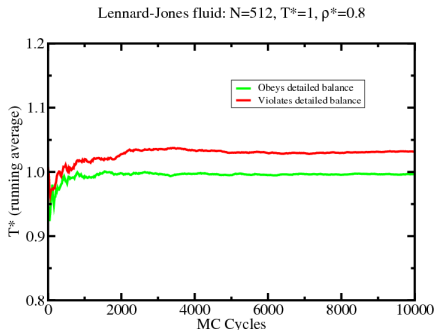
Workshop

After running `mc_nvt`, check that the simulation average of T_{conf} matches the input temperature T .

Reference

Butler, Ayton, Jepps & Evans,
J. Chem. Phys. **109**,
6519 (1998)

Example



The configurational temperature is **sensitive to programming errors**.

- **Green:** correct implementation — T_{conf} converges to input $T^* = 1.0$.
- **Red:** violates detailed balance — T_{conf} converges to a *wrong* value.

LJ fluid: $N = 512$, $T^* = 1$,
 $\rho^* = 0.8$

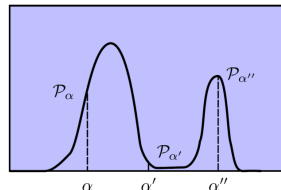
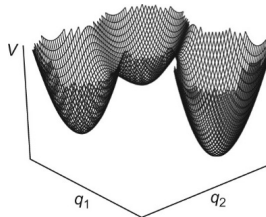
Butler et al., *J. Chem. Phys.* **109**,
6519 (1998)

Ergodicity

Definition

"All areas of configuration space are accessible from any starting point"

- Difficult to prove a priori that a MC algorithm is ergodic.
- Many elementary MC algorithms are *practically* non-ergodic — important regions of configuration space are very difficult to access.
- Common cause: **free energy barriers** (e.g. first-order phase transitions).



Isobaric-isothermal ensemble

simulations

Volume changes

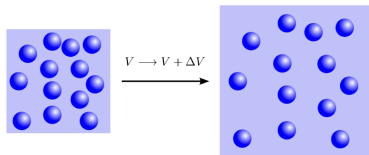
In the NPT ensemble, we also propose **volume changes**:

$$V' = V + \delta V, \quad \delta V \in [-V_{\max}, V_{\max}]$$

Accept with Metropolis criterion for the enthalpy:

$$\Delta H = \Delta U + P\Delta V - Nk_B T \ln\left(\frac{V'}{V}\right)$$

- Particle positions are **scaled** during volume moves $\Rightarrow$ use fractional coordinates.
- $\langle V \rangle$ gives the EOS volume at (T, P) .



Volume move: $V \rightarrow V + \Delta V$.
All positions scale so fractional coordinates are preserved.

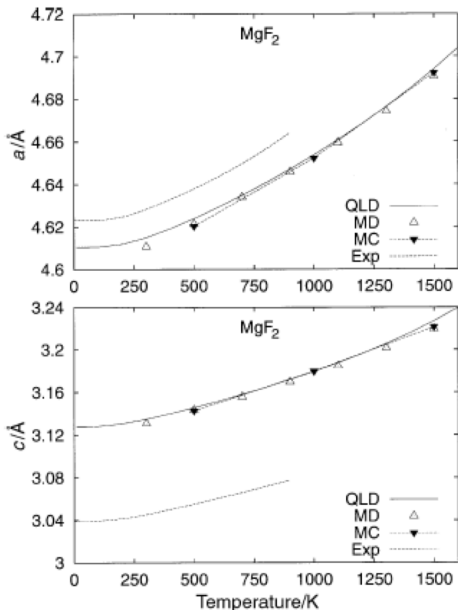
Equations of state from MC

MC calculations (both NVT and NPT) have been extremely useful in establishing equations of state.

MgF₂ — lattice parameters vs. T :

- MC Monte Carlo
- MD Molecular dynamics
- QLD Quasiharmonic Lattice Dynamics

NPT MC and MD agree well with experiment.
QLD (harmonic)



Grand canonical ensemble

In some problems we want to know the **average number of particles** as a function of external conditions.

Example: how much gas is absorbed in a zeolite as a function of the reservoir pressure and temperature?

Grand canonical (μVT) ensemble:
 T , V and μ are fixed.

- The gas reservoir imposes T and μ on the adsorbed molecules.
- The number of particles is allowed to **fluctuate** during the simulation.

Xe in silicalite

A classic GCMC application:

how many Xe atoms adsorb as a function of pressure P and temperature T ?

Activity

$$z = \frac{e^{\mu/k_B T}}{\Lambda^3}$$

plays the role of a fixed external chemical potential.

Grand-canonical Monte Carlo

In addition to particle *moves*, GCMC attempts particle **insertions** and **deletions**:

Insertion ($N \rightarrow N + 1$):

$$P_{\text{acc}} = \min\left(1, \frac{zV}{N+1} e^{-\beta\Delta U}\right)$$

Deletion ($N \rightarrow N - 1$):

$$P_{\text{acc}} = \min\left(1, \frac{N}{zV} e^{-\beta\Delta U}\right)$$

where $z = e^{\beta\mu}/\Lambda^3$ and $\beta = 1/k_{\text{B}}T$.

Grand canonical partition function

$$\Xi = \sum_{N=0}^{\infty} z^N Q_{NVT}$$

Limitation

At high density, insertions are rarely successful.

Alternatives: expanded ensemble, configurational-bias MC.

$\langle N \rangle$ vs. μ gives the **adsorption isotherm** directly.

Ensemble Averages

Assuming we are sampling the equilibrium distribution, ensemble averages are **simple averages over configurations**:

$$\langle A \rangle \approx \frac{1}{M} \sum_{k=1}^M A(\mathbf{r}_k^N)$$

Statistical errors

Successive configurations are *correlated*, so naive error estimates are too small.

Block averaging: divide run into n_b blocks, compute block averages $\bar{A}_b$; error $\approx \sigma(\bar{A}_b)/\sqrt{n_b}$ (block length $\gg$ correlation time τ_0).

Assumption

Statistical errors can be estimated assuming that block averages are **statistically uncorrelated** when the block size exceeds the correlation time.

Key properties: NPT ensemble

Key properties computed from NPT Monte Carlo:

Enthalpy per atom

$$h = \frac{\langle H \rangle}{N} = \frac{\langle U + PV \rangle}{N}$$

Isothermal compressibility

$$\kappa_T = \frac{\langle V^2 \rangle - \langle V \rangle^2}{k_B T \langle V \rangle}$$

Isobaric heat capacity

$$\frac{C_P}{Nk_B} = \frac{\langle H^2 \rangle - \langle H \rangle^2}{N(k_B T)^2}$$

Thermal expansion coefficient

$$\alpha_P = \frac{\langle VH \rangle - \langle V \rangle \langle H \rangle}{k_B T^2 \langle V \rangle}$$

Volume / density

$$\langle V \rangle \Rightarrow \rho = \frac{N}{\langle V \rangle}$$

Chemical potential (Widom)

$$z = \frac{\langle V e^{-\Delta U_{\text{test}}/k_B T} \rangle}{(N+1) \langle V \rangle}$$

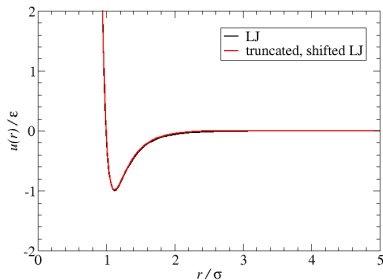
Truncated-shifted Lennard-Jones atoms

The **Lennard-Jones** pair potential:

$$u_{\text{LJ}}(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]$$

Cut-and-shifted ($r_c = 2.5\sigma$):

$$u(r) = \begin{cases} u_{\text{LJ}}(r) - u_{\text{LJ}}(r_c) & r \leq r_c \\ 0 & r > r_c \end{cases}$$



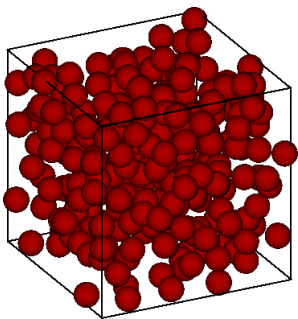
Full LJ (black) vs. truncated, shifted LJ (red). The shift removes the discontinuity at r_c .

Workshop

State point: $T^* = 2.0$, $\rho^* = 0.5$,
 $N = 256$, $r_c = 2.5\sigma$.

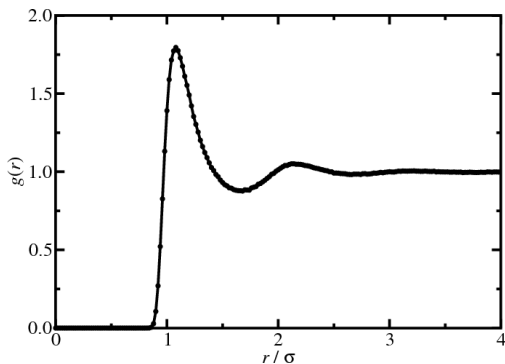
Critical point: $T_c = 1.0779$,
 $\rho_c = 0.3190$.

Results: Lennard-Jones fluid



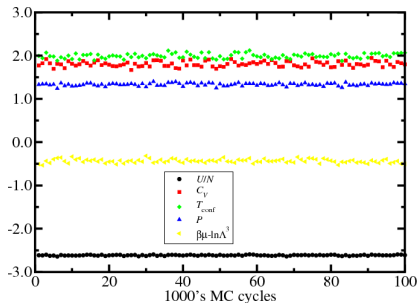
Equilibrated configuration
256 LJ atoms

$$T^* = 2.0, \rho^* = 0.5$$

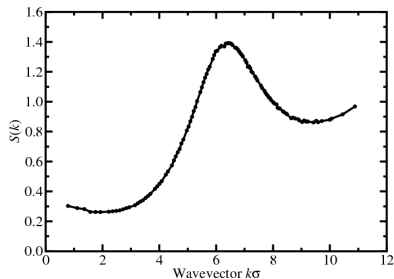


Pair distribution function $g(r)$. Peak at $r \approx 2^{1/6}\sigma$ (first coordination shell); $g(r) \rightarrow 1$ at large r .

Results: Lennard-Jones fluid (2)



Running averages of U/N , C_V , T_{conf} , P and $\beta\mu - \ln\Lambda^3$ over 10^5 MC cycles confirm equilibrium.



Static structure factor $S(k)$ vs. wavevector $k\sigma$. Peak at $k\sigma \approx 6$ corresponds to the first coordination shell.

Workshop

Compare with the fitted EOS (Thol et al. 2015).

Results: Lennard-Jones fluid (3)

Regardless of the ensemble (NVT, NPT or μ VT), at the *same* thermodynamic state the average properties are **nearly identical** (ensemble equivalence in the thermodynamic limit).

Workshop

Run simulations in different ensembles at the same state point and compare the thermodynamic averages.

		NVT $N = 256$ $V = 512\sigma^3$ $T^* = 2$	NPT $N = 256$ $P^* = 1.3337$ $T^* = 2$	μ VT $z^* = 0.6477$ $V = 512\sigma^3$ $T^* = 2$
Property				
Density	$\rho\sigma^3$	0.5	0.49915(38)	0.50119(35)
Temperature	$k_B T_{\text{cont}}/\epsilon$	1.9942(54)	1.9981(44)	1.9948(50)
Energy	$U/N\epsilon$	-2.6147(11)	-2.6105(19)	-2.6173(98)
Enthalpy	$H/N\epsilon$	0.0527(63)	0.0649(38)	0.081(13)
Pressure	$P\sigma^3/\epsilon$	1.3337(31)	<u>1.3325(32)</u>	1.3525(36)
Chemical potential	$\beta\mu - 3\ln(\Lambda/\sigma)$	-0.4343(44)	<u>-0.4478(46)</u>	-0.4356(14)
Activity	$z\sigma^3$	0.6477(28)	<u>0.6390(29)</u>	<u>0.6469(10)</u>
Heat capacity	C_V/Nk_B	1.8077(61)	3.932(57)	
Heat capacity	C_P/Nk_B		0.3147(72)	
Compressibility	$\chi\epsilon/\sigma^3$			0.3121(67)

Comparison of thermodynamic properties in NVT, NPT and μ VT ensembles at the same LJ state point.

Finite size effects

Real simulations use finite N ($\sim 10^2$ – 10^4), introducing systematic deviations from bulk behaviour.

Sources:

- Truncation and PBC alter effective potential
- Long-wavelength fluctuations suppressed when $L < \xi$ (correlation length)
- Enhanced fluctuations vs. thermodynamic limit

Corrections:

- Long-range corrections to U and P
- Finite-size scaling analysis

Long-range correction to energy

$$U_{\text{lr}} = 2\pi N\rho \int_{r_c}^{\infty} u_{\text{LJ}}(r) r^2 dr$$

Long-range correction to pressure

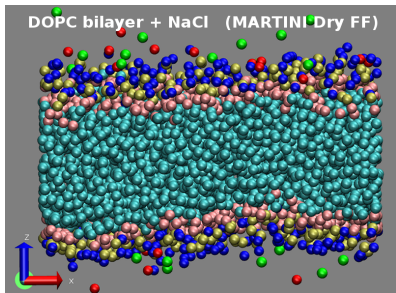
$$P_{\text{lr}} = \frac{2\pi\rho^2}{3} \int_{r_c}^{\infty} r \frac{du_{\text{LJ}}}{dr} r^2 dr$$

DL_MONTE General Purpose Monte Carlo code

Open-source parallel MC
(CCP5).

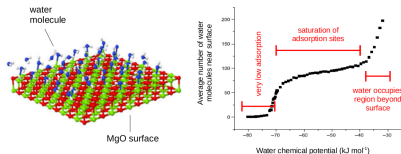
https://www.ccp5.ac.uk/DL_MONTE
Brukhno et al., *Mol. Simul.* **47**, 131
(2021)

Applications:



NPT simulation of a free-standing DOPC
phospholipid bilayer in NaCl electrolyte.
(Courtesy Dr Andrey Brukhno)

NPT simulation of water (SPC/E) at ambient conditions.



GCMC simulation of adsorption of water
molecules onto a MgO surface. (Courtesy
John Purton)

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